Fiscal Year:	FY 2021	**Task Last Updated:**	FY 01/18/2021
PI Name:	Wolverton, Scot Ph.D.		
Project Title:	Characterizing Plant Gravity Perception Systems		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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City:	Delaware	**State:**	OH
Zip Code:	43015-2333	**Congressional District:**	12
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2014 Space Biology Flight (ILSRA) NNH14ZTT002N
Start Date:	03/16/2015	**End Date:**	03/15/2022
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA KSC
Contact Monitor:	Levine, Howard	**Contact Phone:**	321-861-3502
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 3/15/2022 per NSSC information (Ed., 3/16/21) NOTE: Extended to 3/15/2021 per NSSC information (Ed., 12/16/19) NOTE: Extended to 3/15/2020 per NSSC information (Ed., 12/17/18) NOTE: Extended to 3/15/2019 per NSSC information (Ed., 12/18/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX15AG55G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Plant growth is characterized by plasticity, allowing environmental cues like light and gravity to play a significant role in determining the overall form of the plant body. Statolith sedimentation in specialized cells represents one way that plants perceive gravity. Plants lacking functional statoliths show reduced but consistent gravity responses, suggesting the presence of a non-statolith sensory system. This proposal seeks to investigate the gravity perception machinery in roots of several genotypes of the higher plant Arabidopsis thaliana by taking advantage of the unique microgravity environment of Earth orbit on the International Space Station (ISS). One thrust of the work described here is to characterize the physiological properties of the putative non-statolith sensory system, including the threshold acceleration force required to activate it, and its capacity to modulate downstream growth regulation machinery. The second thrust is to determine whether the two gravity sensory systems control a single or multiple output systems. Addressing this question requires the microgravity environment in order to quantify the kinetics of gravitropism in roots having or lacking functional statoliths under fractional g conditions. The overall outcome of the work proposed here will be an improved understanding of how higher plants perceive gravity and transduce this physical cue into directional growth, a fundamental requirement for improving plants for production on Earth and in long-term spaceflight missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The overall outcome of the work proposed here will be an improved understanding of how higher plants perceive gravity and transduce this physical cue into directional growth, a fundamental requirement for improving plants for production on Earth and in long-term spaceflight missions. In particular, the differences in growth responses and gene regulation between wild-type and starchless mutants will help elucidate the contributions of statolith-based sensing to plant gravity responses.
Task Progress:	The funding period March 2020 — March 2021 was one characterized by interruptions and delays due to the SARS-CoV2 pandemic. Much of the work planned for the year was significantly delayed due to closures and shutdown orders implemented by the State of Ohio to help control the spread of the virus. The three objectives for the year included the completion of sequencing on samples from the Plant Gravity Perception (PGP) flight experiments, bioinformatic analysis of the RNA-seq data arising from that sequencing, as well as completion of ground-based controls and mutant analysis. Progress on each of these objectives is described below. Samples returned from flight, where they were exposed for 36 h to specific artificial gravity treatments using the European Modular Cultivation System (EMCS) centrifuge, were processed to extract RNA. Seedlings were dissected into 4 organs (cotyledons, hypocotyls, mature roots, and root tips) and RNA was extracted separately for each organ across a range of 12 gravity treatments for both WT and starchless (pgm-1) mutants. Each organ-gravity-genotype sample was replicated 4 times in the design to maximize statistical robustness. After confirming the quality of RNA samples via Bioanalyzer chip, they were shipped to a core facility for consistent library preparation using the Takara low-input library kit. Libraries were sequenced on an Illumina NovaSeq 6000 using paired-end, 150 bp reads. Both the library preparations and the sequencing were delayed due to laboratory shutdowns, with the final flight reads completed in November 2020. Due to the delays in completing the sequencing process, we have only recently started to analyze the RNA-seq data, which totals 400 sample libraries and approximately 2 TB of data. We are coordinating the upload of the raw read files to NASA's GeneLab platform for widespread availability of the data and analysis with the community. Fortunately, our collaborators, Sarah Wyatt and Al Meyers, have built and tested the analysis pipeline to be used for processing the data, using our previous pilot data set for development and optimization. We anticipate being able to complete the first analysis of these data by Summer 2021. These data will allow us to directly probe the effect of gravity as a continuum on plant seedling growth and establishment from a minimum of 0.006 g up to 1.0 g, representing the most complete and systematic look at the effect of this key variable on gene expression to date. In addition to the work on preparing and sequencing flight samples, we also made progress on identifying novel components of the gravity signaling pathway in roots. We had previously identified a list of 124 genes showing contrasting expression patterns between WT and pgm-1 mutants under vertical compared to gravistimulated conditions. As a first step toward testing the roles of some of these genes in gravity response, students tested about 35 T-DNA lines with insertions in some of these genes. While the pandemic and ensuing shutdown precluded students from working in the lab over the summer, students made progress in Fall 2020 toward confirming the genotypes and transcript abundance in these mutant lines using RT-qPCR.
Bibliography Type:	Description: (Last Updated: 05/04/2026)
Abstracts for Journals and Proceedings	Meyers A, Wolverton C, Wyatt S. "Leveraging Molecular Data from Fractional Gravity Environments on Earth and ISS." Gravity Perception and Response in Plants and Fungi: Ground and Space Studies. Committee on Space Research (COSPAR) 2021-Hybrid, 43rd Scientific Assembly, Sydney, Australia, January 28-February 4, 2021. Committee on Space Research (COSPAR) 2021-Hybrid, 43rd Scientific Assembly, Sydney, Australia, January 28-February 4, 2021. , Jan-2021
Abstracts for Journals and Proceedings	Wolverton C, Meyers A, Wyatt S. "Searching for Components of the Non-Statolith Gravity Sensing System in Plants." Gravity Perception and Response in Plants and Fungi: Ground and Space Studies. Committee on Space Research (COSPAR) 2021-Hybrid, 43rd Scientific Assembly, Sydney, Australia, January 28-February 4, 2021. Committee on Space Research (COSPAR) 2021-Hybrid, 43rd Scientific Assembly, Sydney, Australia, January 28-February 4, 2021. , Jan-2021
Articles in Peer-reviewed Journals	Overbey EG, Saravia-Butler AM, Zhang Z, Rathi KS, Fogle H, da Silveira WA, Barker RJ, Bass JJ, Beheshti A, Berrios DC, Blaber EA, Cekanaviciute E, Costa HA, Davin LB, Fisch KM, Gebre SG, Geniza M, Gilbert R, Gilroy S, Hardiman G, Herranz R, Kidane YH, Kruse CPS, Lee MD, Liefeld T, Lewis NG, McDonald JT, Meller R, Mishra T, Perera IY, Ray S, Reinsch SS, Rosenthal SB, Strong M, Szewczyk NJ, Tahimic CGT, Taylor DM, Vandenbrink JP, Villacampa A, Weging S, Wolverton C, Wyatt SE, Zea L, Costes SV, Galazka JM. "NASA GeneLab RNA-seq consensus pipeline: Standardized processing of short-read RNA-seq data." iScience. 2021 Apr 23;24(4):102361. https://doi.org/10.1016/j.isci.2021.102361 ; PMID: 33870146; PMCID: PMC8044432 , Apr-2021